

Design of an FPGA-Based Hardware Accelerator for Low-Latency Neural Network Inference

Experiment 4 - Intermediate

Vivado 2025.1.1 | xc7z020clg400-1 | 50 MHz | audit: 9 October 2026 (IST)

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1. Objective

Implement a trained 4-input, 4-hidden-neuron, 3-output Iris classifier on the PYNQ-Z2. Compare a pipelined parallel accelerator with a sequential shared-MAC core using identical frozen weights, arithmetic, inputs and clock frequency.

The engineering question is whether parallel hardware reduces inference latency, and what DSP/resource cost that reduction requires. Software performs training and input preparation; neural inference itself executes in FPGA logic.

Verified metric	Result
Physical numeric correctness	45/45 held-out inputs match every frozen integer score
Classification accuracy	43/45 (95.56%); two model errors
Core latency at 50 MHz	Parallel: 4 cycles / 80 ns; sequential: 36 cycles / 720 ns
Architecture comparison	9x lower core latency; host/JTAG transfer time excluded
Routed timing	Setup +5.813 ns; hold +0.048 ns; no failing endpoints

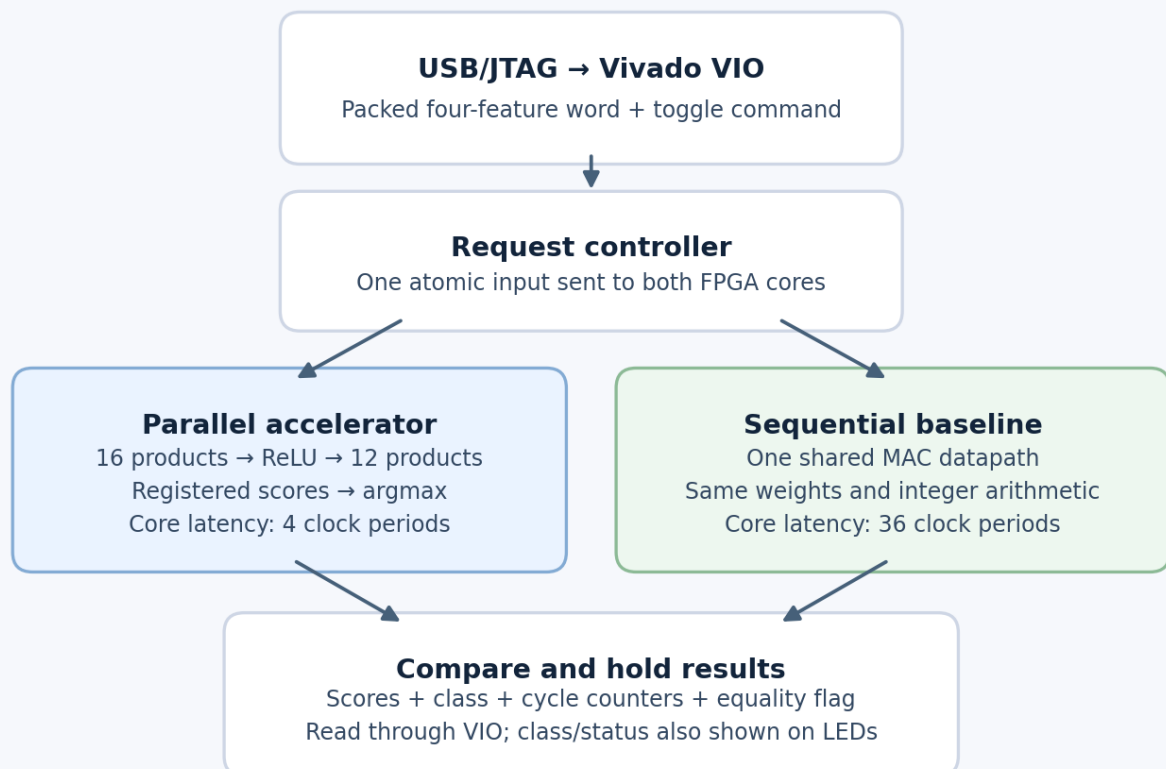
Demonstration scope

This is a small classification benchmark demonstrating fixed-point inference and an FPGA architecture tradeoff. It does not validate an application-specific sensor system or claim an end-to-end CPU speedup. The VIO transaction wrapper runs one comparison at a time; the parallel core alone accepts a new input each clock.

2. Block Diagram

Project 4 • Trained 4 → 4 → 3 neural network

PYNQ-Z2 / SystemVerilog / equal-clock FPGA architecture comparison



50 MHz target: 80 ns vs 720 ns after timing closure and board verification. JTAG overhead excluded.

Source-derived architecture diagram from the original implementation pack. Clocking Wizard generates 50 MHz from the board 125 MHz PL clock; lock release is synchronized. Vivado VIO drives reset, a command toggle and the packed input word. The controller launches both cores and holds their results for JTAG inspection.

Model and numeric representation

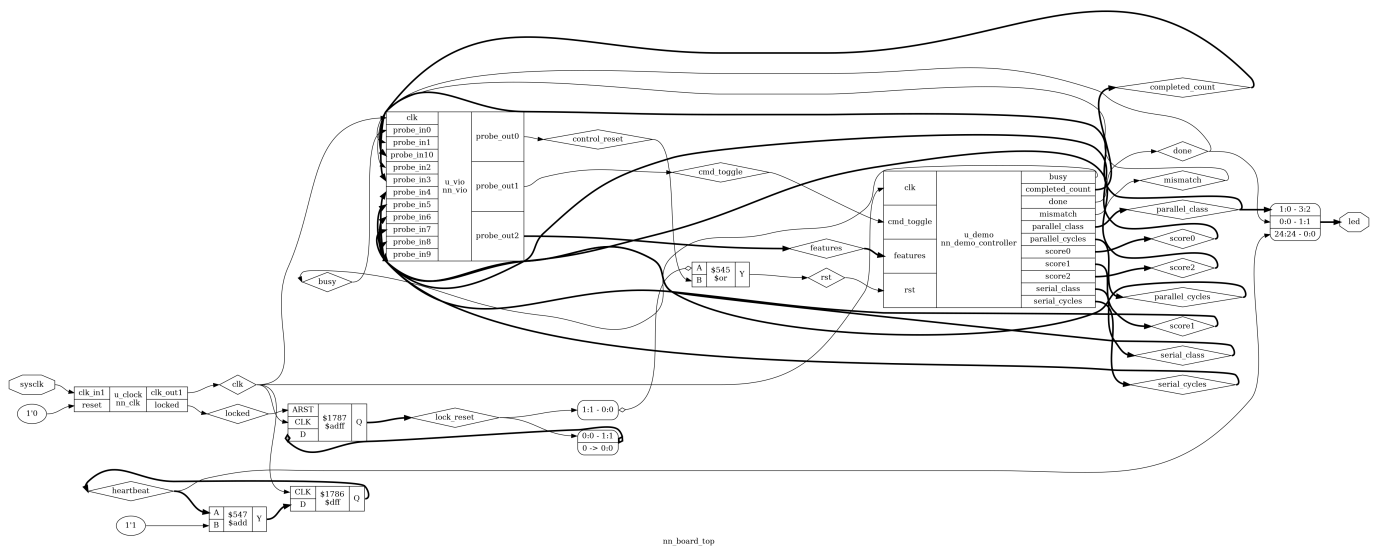
Item	Frozen implementation
Dataset split	90 training / 15 validation / 45 held-out; 150 Iris records
Features and classes	Sepal length/width and petal length/width; 0 Setosa, 1 Versicolor, 2 Virginica
Input and weight formats	Signed 8-bit inputs at scale 32; signed 8-bit weights at scale 64
Internal arithmetic	16-bit hidden activations; 32-bit sums/scores; input byte i is features[8*i + : 8]
Activation and prediction	Arithmetic shift by 6 then ReLU saturation to 0..32767; signed argmax with lower-index tie priority

3. RTL Design

File / instance	Function
nn_board_top.sv	Board clock/reset, clock/VIO IP, heartbeat and class/status LEDs
nn_demo_controller.sv / u_demo	Toggle launch, busy rejection, sticky scores/classes, counters and mismatch comparison
nn_parallel.sv / u_parallel	16 first-layer products, ReLU, 12 second-layer products, scores and argmax
nn_serial.sv / u_serial	One shared multiplication/MAC datapath controlled by a sequencer
nn_weights.vh; nn_math.vh	Frozen weights/biases; signed extension, saturation and argmax helpers
pynq_z2.xdc	Board sysclk/LED pins and exceptions; primary clock supplied by Clocking Wizard

Parallel pipeline: elapsed periods from acceptance

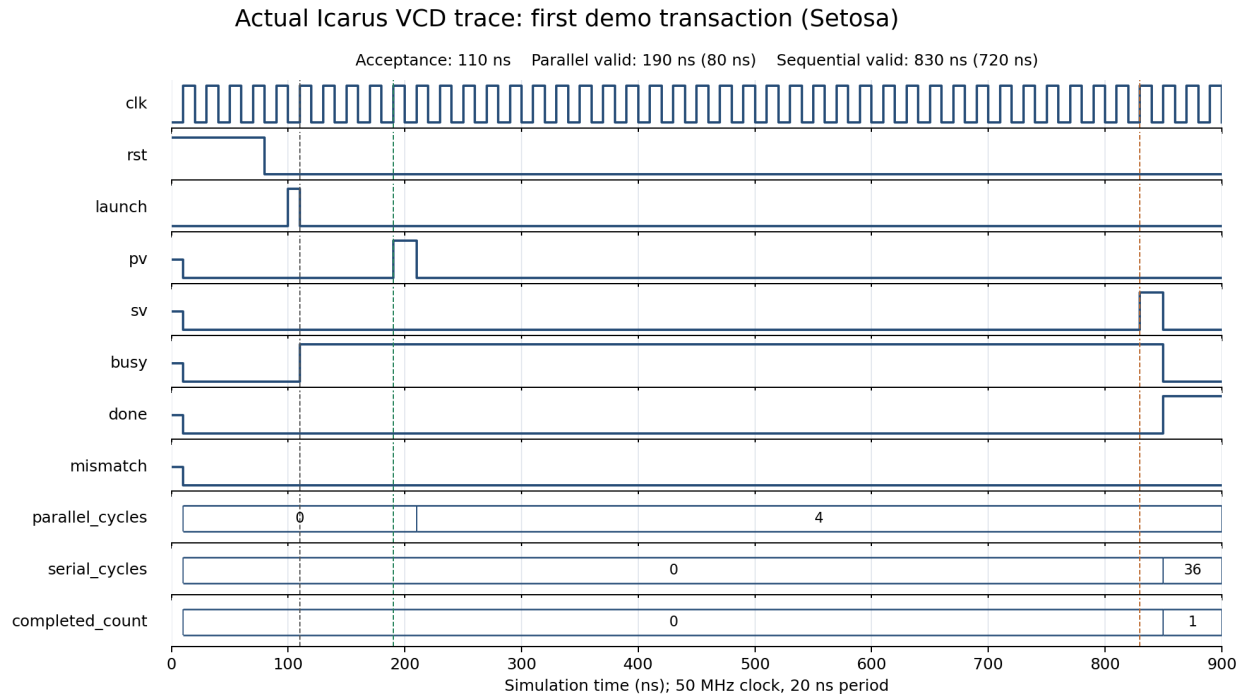
E0: input/products captured. **E1:** first-layer sums, bias, rescale and ReLU. **E2:** second-layer products. **E3:** three scores. **E4:** valid scores and class. Five registered stages span four elapsed clock periods.



Actual source-elaborated schematic generated with Yosys 0.33 and Graphviz. Clocking Wizard and VIO are port-accurate black boxes for this view; their vendor internals are not simulated by these declarations. This is not a Vivado GUI capture. Full-resolution PNG/SVG, netlist, tool transcript and reproduction recipe are included.

4. Simulation Results

Original local XSim evidence is preserved in Results/vivado_batch.log and simulation_pass.flag. Icarus Verilog 12.0 regression and VCD capture provide additional simulation evidence. On 9 October 2026 (IST), the unchanged source passed the full suite and three-case waveform bench again in local Vivado XSim 2025.1.1. The new outputs are current_xsim_regression.txt and current_xsim_waveform.txt; the original Icarus waveform is retained.



Measured from actual VCD transitions: first input accepted at 110 ns; parallel output valid at 190 ns; sequential output valid at 830 ns. Sticky controller outputs are captured later, so the picture distinguishes core valid timing from the host-visible comparison result.

PASS reset abort/recovery, argmax ties, ReLU saturation
PASS parallel: 4502 vectors, full-rate bursts + bubbles, latency=4 periods
PASS wrapper: 150 Iris samples, held toggle, busy rejection, reset abort, counters
PASS serial: 4502 vectors, input latching, latency=36 periods
ALL TESTS PASSED
Testbench/tb_nn.sv:169: \$finish called at 3421600000 (1ps)

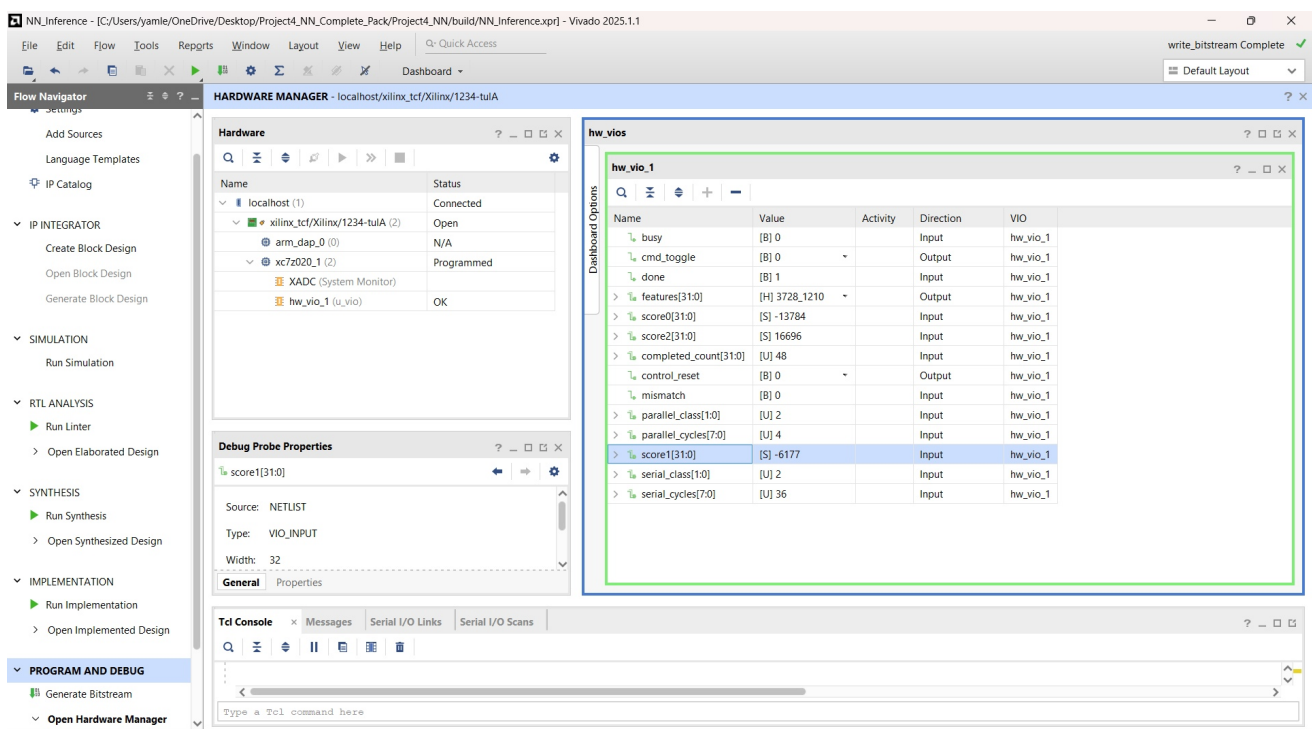
The full regression verifies exact scores and classes for 4,502 vectors per core, full-rate parallel bursts and bubbles, sequential input latching, 150 controller cases, held-toggle non-retriggering, busy rejection, reset cancellation/recovery, argmax ties and ReLU saturation. The supplementary waveform bench independently checks the three recorded demo inputs and 4/36-cycle valid timing.

5. Hardware Implementation

The team programmed the physical PYNQ-Z2 over USB/JTAG and ran `hardware_test.tcl` on 7 October. The script checked 45 held-out records against frozen reference scores. Board photo, hardware PASS screenshot, three-case console output and final VIO screenshot are preserved in Images/.

Scope	LUTs	FFs	DSPs
Parallel core	247	263	28
Sequential core	208	330	1
Whole board design	1515	2648	29

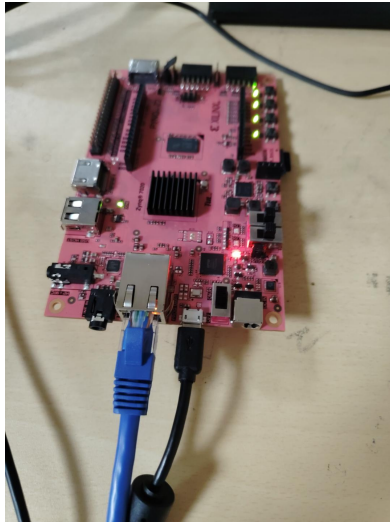
Raw routed reports: setup +5.813 ns; hold +0.048 ns; pulse width +2.000 ns; all four bus-skew checks pass, minimum slack +19.066 ns. The original matching 6 October bitstream and LTX pair are included; the reports describe the recorded build.



Final VIO output for 37281210: both classes 2, scores -13784 / -6177 / 16696, mismatch 0, cycles 4/36, completed count 48. The visible values were read from actual FPGA probes.

Numeric correctness is 45/45, while species accuracy is 43/45. Records 68 and 138 are misclassified by the model but their FPGA scores match the software model exactly. The saved CSV was independently cross-checked against all frozen test records.

5. Hardware Implementation (continued)



PYNQ-Z2 setup photograph: PYNQ-Z2 with USB/JTAG and Ethernet attached. Ethernet/Jupyter is not used for this RTL/VIO inference workflow. VIO availability and the heartbeat were observed after programming.

Warnings and script maintenance

DRC reports 44 warnings and no errors; methodology reports four LUTAR-1 reset warnings inside AMD debug-hub IP. DSP pipeline warnings are performance recommendations; timing meets the chosen 50 MHz target. ZPS7-1 remains recorded in the PL-only design. Raw reports are retained, and successful physical VIO testing supplies board/clock evidence for this setup.

The VIO script originally parsed all readbacks as hexadecimal. Decimal display caused a false timeout although hardware finished correctly. `nn_hw_read` now uses `INPUT_VALUE_RADIX` and restores signed scores as 32-bit two's-complement values. The corrected reader was tested in the live session; the persistent repository script includes it. HDL/model and bitstream are unchanged.

6. Applications

The architecture illustrates low-latency embedded classification, FPGA ML teaching and controlled resource/latency comparisons. Applying it to industrial inspection or sensor processing requires suitable training data, preprocessing and domain validation. More parallel arithmetic trades DSP use for latency; the small Iris model makes that tradeoff easy to reproduce.

7. Conclusion

A trained quantized network executes correctly in two FPGA architectures. Both match the integer reference; the parallel implementation reduces core latency from 36 to 4 cycles at the same 50 MHz clock. The experiment demonstrates fixed-point arithmetic, pipelining, resource sharing, verification and hardware debugging.

Future scope: larger datasets/networks, configurable weights, measured power and an AXI-stream/DMA interface. Team identity and final demo URL are recorded.

Evidence/RAW_VIDEO.md preserves the raw recording; Video_Link.txt points to the final edit. Content and public reviewer access need review (Documentation/VIDEO_REVIEW.md). The final video must include introduction, problem, architecture/RTL presentation, working hardware and conclusion.

The original native Vivado project descriptor and clock/VIO IP configuration files were recovered from the laptop and archived in `FPGA_Project/Project4_NN_Vivado_Source_Project.zip` with the source tree. Original descriptor/IP bytes are preserved, including machine-specific paths and historical run settings. Generated caches/runs are excluded. Use the supplied project-generation Tcl for a clean relocated rebuild.

Evidence sources: `Data/model.json` and `iris_cases.csv`; `Results/*.rpt` and `vivado_batch.log`; `Evidence/hardware_test_results.csv` and `hardware_validation.json`; `Simulation/transcript.txt` and `waveform.vcd`. Dataset attribution and original guide are included in `Data/ATTRIBUTION.md` and `START_HERE.md`.